

RAYONG, Thailand

WMO: 484780

Lat: 12.6339N

Lon: 101.3400E

Elev: 15

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	67.1	69.4	53.9	61.7	74.0	57.3	70.0	76.0	12.2	84.6	11.7	85.0	2.8	350	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.6	95.3	82.3	93.6	81.4	92.5	80.9	83.9	92.4	83.3	91.1	82.6	89.9	4.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.9	166.1	89.1	81.0	161.3	88.2	80.7	159.4	87.9	48.1	92.3	47.2	91.2	46.6	90.3	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	11.9	10.8	63.2	96.8	2.8	3.4	61.2	99.2	59.6	101.2	58.0	103.2	55.9	105.6
			57.4	85.0	3.2	1.8	55.1	86.3	53.2	87.4	51.4	88.4	49.0	89.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.6	80.1	82.6	84.8	86.6	86.1	85.4	84.7	84.4	83.3	82.9	82.4	80.4	(6)
(7)		DBStd	3.18	2.85	2.87	2.81	2.80	2.50	2.20	2.18	1.95	2.05	1.82	2.09	3.12	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12278	932	913	1079	1098	1120	1062	1075	1065	1000	1019	972	943	(10)
(11)		CDD65	6803	467	493	614	648	655	612	610	600	550	554	522	478	(11)
(12)		CDH74	81383	4679	5555	7605	8658	8792	8044	7742	7501	6483	5979	5557	4788	(12)
(13)		CDH80	35200	1690	2150	3420	4482	4425	3818	3396	3165	2468	2148	2181	1857	(13)
(14)	Wind	WSAvg	4.0	3.1	3.9	4.1	4.0	4.3	5.3	5.4	5.3	3.8	2.6	3.2	3.4	(14)
(15)	Precipitation	PrecAvg	58.1	1.1	1.2	2.7	3.1	7.5	7.4	7.3	5.7	11.0	8.3	1.9	0.3	(15)
(16)		PrecMax	77.7	4.4	4.4	11.2	8.9	15.7	13.3	13.5	13.5	18.9	15.9	4.4	1.5	(16)
(17)		PrecMin	38.9	0.0	0.0	0.0	0.1	2.6	0.5	1.8	1.9	4.5	2.4	0.0	0.0	(17)
(18)		PrecStd	9.4	1.3	1.0	2.4	2.1	3.5	3.3	3.9	2.6	3.3	2.9	1.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.8	94.3	96.1	98.0	96.3	95.2	94.9	93.6	93.2	93.5	93.6	94.2	(19)
(20)			MCWB	79.3	81.2	81.8	84.1	83.3	82.1	82.4	81.3	81.4	80.9	79.7	79.5	(20)
(21)		2%	DB	90.3	91.6	93.5	95.0	93.6	92.7	91.8	91.3	90.8	91.4	91.8	91.5	(21)
(22)			MCWB	77.0	79.0	81.4	82.6	82.4	81.4	81.0	80.3	80.5	79.9	78.4	77.2	(22)
(23)		5%	DB	88.8	89.7	91.4	93.0	92.0	90.9	89.8	89.5	89.2	89.8	90.4	89.7	(23)
(24)			MCWB	76.4	78.2	80.6	82.0	81.9	81.0	80.3	80.1	80.0	79.4	77.8	76.3	(24)
(25)		10%	DB	87.3	88.0	89.5	91.3	90.6	89.4	88.2	87.9	87.5	88.2	89.1	87.9	(25)
(26)			MCWB	76.0	78.1	80.2	81.7	81.5	80.7	80.0	79.9	79.7	79.2	77.3	75.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	82.8	84.1	85.0	85.0	83.9	83.7	83.3	83.0	83.2	82.5	81.7	(27)
(28)			MCD B	89.9	91.6	93.5	95.3	94.0	91.3	91.5	90.3	89.8	90.5	90.8	91.1	(28)
(29)		2%	WB	80.1	81.4	82.6	83.7	83.6	82.8	82.6	81.9	81.8	81.8	81.0	80.0	(29)
(30)			MCD B	87.0	89.1	91.1	92.6	91.5	89.5	88.9	88.1	87.9	88.6	88.8	88.3	(30)
(31)		5%	WB	79.3	80.4	81.7	82.9	82.9	82.2	81.8	81.1	81.0	80.9	80.1	78.9	(31)
(32)			MCD B	85.7	87.2	89.3	91.0	90.1	88.5	87.7	87.0	86.8	87.3	87.4	86.5	(32)
(33)		10%	WB	78.5	79.6	80.9	82.2	82.2	81.6	81.0	80.6	80.3	80.1	79.2	77.9	(33)
(34)			MCD B	84.5	86.0	87.9	89.7	88.9	87.8	86.8	86.3	85.9	86.1	85.9	85.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.5	12.3	10.6	10.6	9.6	8.7	8.4	8.2	9.4	11.7	14.3	16.2	(35)
(36)			MCDBR	17.6	14.9	13.7	13.2	12.0	11.4	11.5	11.1	12.0	14.5	16.3	17.8	(36)
(37)			MCWBR	8.5	6.8	5.7	5.4	5.4	5.3	5.2	5.1	5.4	6.4	7.2	8.3	(37)
(38)		5% WB	MCD BR	14.6	12.3	12.0	12.0	11.0	10.2	10.2	9.8	10.7	12.9	14.7	15.9	(38)
(39)			MCWBR	7.2	5.7	5.3	5.2	5.3	5.2	5.2	5.0	5.3	6.2	6.9	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.544	0.594	0.600	0.541	0.478	0.463	0.461	0.461	0.469	0.527	0.489	0.478	(40)	
(41)		taud	1.992	1.872	1.907	2.084	2.311	2.373	2.370	2.361	2.321	2.124	2.216	2.211	(41)	
(42)		Ebn at Noon	242	235	237	250	262	263	264	267	267	250	256	257	(42)	
(43)		Edh at Noon	56	65	64	54	42	39	39	40	42	50	44	44	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1599	1717	1758	1835	1618	1562	1511	1552	1499	1519	1586	1576	(44)	
(45)		RadStd	107	92	125	122	103	87	95	105	92	109	117	81	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	-2.38	-0.58	N/A	N/A	N/A	-331	-331	
(47)	Regional (24 neighbors)	N/A	N/A	N/A	+0.54	N/A	+0.32	N/A	N/A	+130	+130	(47)

Nomenclature: See separate page